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# PALEOMAGNETISM OF THE EARLY MIOCENE KANI GROUP IN SOUTHWEST JAPAN AND ITS IMPLICATION FOR THE OPENING OF THE JAPAN SEA

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*Abstract.* Paleomagnetic directions of a volcanic and sedimentary sequence of the Kani Basin, ranging from 22 to 16 Ma, show a clockwise declination shift of about 50°. This result implies that detectable rotation of Southwest Japan did not occur in this period, but after 16 Ma. The initial opening of the Japan Sea before 16 Ma might have caused a southeastward translation of Southwest Japan, which was then followed by clockwise rotation between 16 and 14 Ma.

## Introduction

The Early Miocene to early Middle Miocene sequences distributed in inland basins of Southwest Japan are collectively called the Setouchi Miocene Series. Paleomagnetic directions of the Setouchi Miocene Series are characterized by a clockwise declination shift. In the Mizunami area (Figure 1), the late Early Miocene members of the Setouchi Series showed deflections exceeding 50°, whereas the early Middle Miocene member showed a declination shift less than 30° [Hayashida, 1986]. Deflected paleomagnetic directions have also been observed in the Setouchi Miocene Series in the Ichishi and Morozaki area [Hayashida and Ito, 1984; Hayashida, 1986] and in volcanic and plutonic rocks of the San'in area [Otofuji and Matsuda, 1983; 1984] (Figure 1). The clockwise deflection of the paleomagnetic directions has been explained by rotation of Southwest Japan relative to Eurasia due to back-arc opening of the Japan Sea. Compilation of paleomagnetic and chronological data of Southwest Japan [Otofuji et al., 1985] suggested that the southwestern part of the Japan Sea was opened at about 15 Ma. There is some difficulty, however, in assuming that the whole of the Japan Sea was formed during a short period around 15 Ma as determined from the paleomagnetic data [Kono, 1987]. Other lines of evidence from the Japan Sea basins suggest that the opening started prior to 15 Ma. Tamaki [1986] showed that sediment stratigraphy, basement depth and heat flow data from the Japan and Yamato basins favor an older age ranging from 30 to 15 Ma. Radiometric ages of volcanic rocks dredged from the Yamato Seamount chain range from about 17 to 11 Ma [Kaneoka et al., 1990]. Recently, Legs 127 and 128 of the Ocean Drilling Program (ODP) succeeded in penetrating thick sediments at four sites in the Japan Sea, and provided preliminary micropaleontological data suggesting an age of the sediment/basalt contacts between 14 and 19 Ma [Ingle et al., 1990; Tamaki et al., 1990]. This result suggests that the Yamato basin started rifting in the middle Early Miocene and widened in the late Early to early Middle Miocene. The widening of the Yamato Basin in this period seems in essence concordant with the paleomagnetic evidence from Southwest Japan, but the ODP result also implies that the Sea of Japan had existed to some extent before 15 Ma.

The purpose of this work is to assess the possible rotation of Southwest Japan in the period from 20 to 16 Ma. We report paleomagnetic results from the Setouchi Miocene Series of the Kani Basin, known as the Kani Group, west of the Mizunami basin (Figure 1). The Kani Group includes lacustrine deposits, correlative to the lower part of the Mizunami Group, and underlying volcanic rocks, the Hachiya Formation, which was dated by K-Ar methods at about 22 to 20 Ma [Torii, 1982; Nomura, 1986]. Volcanic products of this age are unique in the Setouchi Miocene Series.

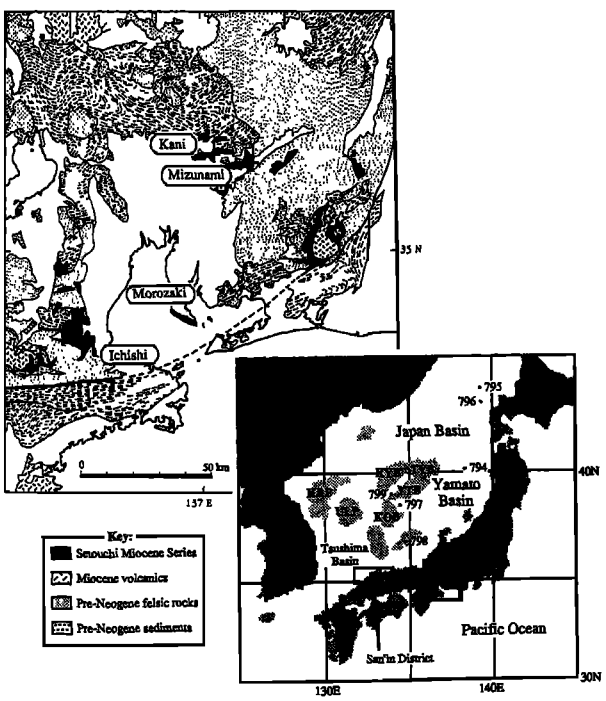



Fig. 1. Map showing distribution of the Setouchi Miocene Series (modified from Geological Survey of Japan, 1982). Index map shows some topographic features in the Japan Sea (YTB: Yamato Bank, KYB: Kita-Yamato Bank, TYB: Takuyo Bank, KOB: Kitaoki Bank, KRP: Korean Plateau, ULP: Ullung Plateau), and ODP Sites 794 to 799. Location of the San'in district [Otofuji et al., 1991] is also shown.

## Field Sampling and Laboratory Procedure

The Kani Group is divided into three formations: the Hachiya, Nakamura and Hiramaki formations in ascending order [Itoigawa, 1980] (Figure 2). The Hachiya Formation is a pile of rhyolitic welded tuffs, brecciated lavas or tuff breccia of andesite or basalt composition, and tuffaceous sediments [Nomura, 1986]. The rhyolitic welded tuffs in the lowermost part yielded a K-Ar date of 22.2 Ma [Torii, 1982]. Andesitic lavas have an age of 20.6±2.0 Ma and 19.8±2.1 Ma [Nomura, 1986]. The Nakamura and Hiramaki formations are composed of lacustrine deposits including some lignite layers

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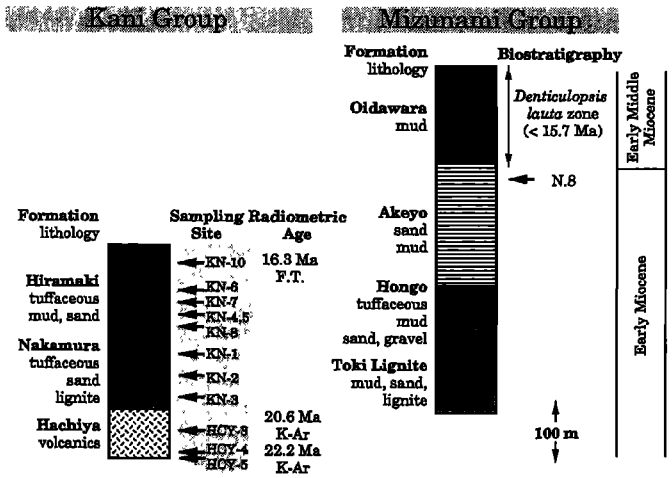



Fig. 2. Stratigraphic summary of the Setouchi Miocene Series in the Kani and Mizunami basins [Itoigawa, 1980]. Also shown are horizons of our paleomagnetic samples and radiometric dates [Torii, 1982; Nomura, 1986; Kobayashi, 1989] of the Kani Group, and biostratigraphic data [Koizumi, 1981; Ibaraki, 1981] of the Mizunami Group.

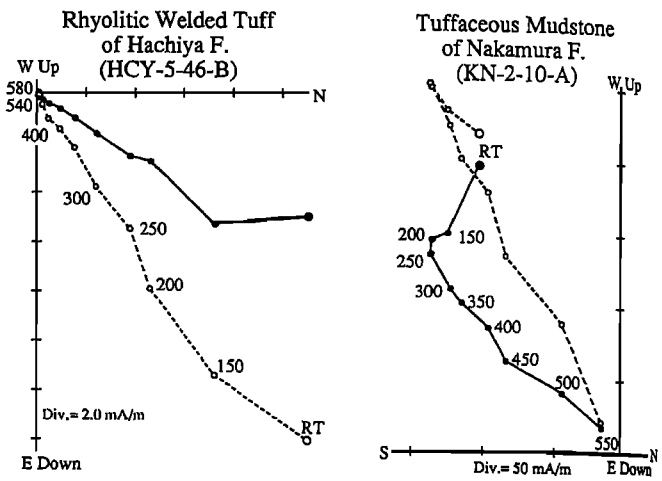



Fig. 3. Examples of stepwise thermal demagnetization data from the Hachiya and Nakamura formations of the Kani Group. Closed and open circles represent magnetization vector end-points projected on the horizontal and vertical planes, respectively, before tilt correction. Demagnetization temperatures are given in degree Celsius for some points.

and tuffaceous sediments. Fission track dates were reported for zircon crystals from pumiceous or fine volcanic ashes of the Nakamura and Hiramaki formations, ranging from 21.3 to 16.3 Ma [Kobayashi, 1989]. The Kani Group is about 300 m thick in total, and tilt of the strata is generally less than 10°.

We collected samples from volcanic rocks of the Hachiya Formation at 3 sites, and tuffaceous sediments of the Nakamura Formation at 3 sites and of the Hiramaki Formation at 6 sites (Figure 2). At each site, 10 to 6 block samples were oriented using a magnetic compass. Generally 2 specimens of cylindrical shape were obtained from each block sample.

Measurement of remanent magnetization was made using cryogenic magnetometers (ScT C-112 at Kyoto University and HOXAN HSR-3RF at Doshisha University). We also used a spinner magnetometer (Schonstedt SSM-1A) for some specimens having magnetization intensity greater than 1 A/m. Magnetic stability was examined by stepwise alternating field (AF) and thermal demagnetization.

## Results

Two magnetic components were detected in most of the volcanic and tuffaceous samples (6 sites) from the Hachiya and Nakamura formations. As shown in the examples of stepwise demagnetization in Figure 3, the magnetic vectors linearly decayed toward the origin during progressive thermal demagnetization between 250 and 550°C. The same component was also observed in the stepwise AF demagnetization over 40 mT. The high temperature or high coercivity components have characteristic directions of northeast-southwest trend, mostly showing low within-site dispersion ($\alpha_{95}$ about 10° or less). We assumed that this stable component represents primary remanent magnetization and that the low temperature/coercivity component originated probably from viscous remanent magnetization. Samples from a brecciated lava of the Hachiya Formation (Site HCY3, dated at 20.6 Ma) had essentially a single stable component, but only showed dispersed directions within the site ($\alpha_{95}$ = 34°). This result can be explained by fragmentation after the cooling below the major blocking temperature range.

Among 6 sites of the Hiramaki Formation, only 2 sites yielded stable magnetization similar to the high temperature component of the Nakamura Formation (Figure 3). Magnetizations of the other 4 sites, collected from two pumiceous tuff layers, were reduced to less than 25% of the original remanence (about 1.0 x $10^{-2}$ A/m) by thermal demagnetization at 200°C, and suffered from unstable viscous overprinting after 500°C. While some linear trends of magnetization vector trajectory were observed between 200 and 400°C, they did not show directional consistency within a site.

Site mean directions from the sedimentary rocks were corrected for bedding (less than 10°) , and those from the welded tuffs were also corrected using their eutaxitic structure (less than 5°) . Tilt correction was not made for the andesite lava; its post-cooling tilt was not clear, while the underlying sediments suggest that it is less than 10°. These data are listed in Table 1, and also plotted on the equal-area projection in Figure 4. Discarding the 2 sites with $\alpha_{95}$ greater than 30° and ambiguous tilt correction, the 6 sites give a total mean (of normal polarity): Declination = 49.9° and Inclination = 53.2° with $\alpha_{95}$ = 9.7°. Our data were obtained from high temperature magnetization components in various lithologies, corrected for paleo-horizontal position. They include both normal and reversed polarities. These facts support the assumption that our data are resulting from primary magnetizations and represent the averaged geomagnetic field direction of the Early Miocene. We could not find significant differences among the site mean data of the Kani Group ranging from 22 to 16 Ma, and also between the means of the Kani Group and the Mizunami Group mostly dated 17 to 16 Ma ([Hayashida, 1986], Figure 4).

Table 1. Paleomagnetic directions of the Kani Group.

| Site | Lithology | Demagnetization | D | I | $\alpha_{95}$ | k | N |
|---|---|---|---|---|---|---|---|
| KN 10 | pumiceous tuff | TH200°C | -118.4 | -37.7 | 14.7 | 40.1 | 4 |
| KN 8 | fine tuff | TH250°C | 44.8 | 59.1 | 13.4 | 26.1 | 6 |
| KN 1 | mudstone | TH400°C | -96.3 | -37.9 | 34.4 | 5.9 | 5 |
| KN 2 | tuffaceous mud | TH350°C | -138.2 | -52.7 | 8.6 | 42.7 | 8 |
| KN 3 | tuffaceous mud | TH400°C | -118.6 | -48.1 | 5.9 | 130.6 | 6 |
| HCY 3 | andesite lava | AF40 mT | -104.8 | -38.0 | 34.1 | 6.0 | 5 |
| HCY 5 | welded tuff | TH150°C | 28.7 | 57.6 | 7.5 | 81.4 | 6 |
| HCY 4 | welded tuff | TH150°C | -126.2 | -60.5 | 12.5 | 29.7 | 6 |

D and I: mean declination and inclination in degree (after tilt correction); $\alpha_{95}$: radius of 95% confidence circle; k: Fisher (1953)'s precision parameter; N: Number of specimens; TH and AF: Thermal and AF demagnetization.

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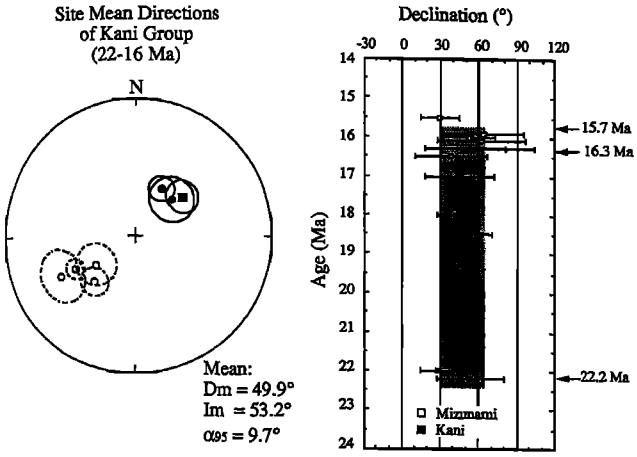



Fig. 4. Site mean paleomagnetic directions of the Kani Group as shown on the equal-area projection (left) and as plots of declination vs. age (right). In the left diagram, open and solid symbols are on the upper and lower hemisphere, respectively, and the solid square indicates the mean direction of the 6 sites of the Kani Group. Paleomagnetic data from the Mizunami Group [Hayashida, 1986] are also shown on the right.

## Discussion

The clockwise declination shift observed in the Mizunami and Kani area is explained by rotation of Southwest Japan relative to Eurasia [Hayashida, 1986; Otofuji et al., 1985]. The Early Miocene poles obtained from Eurasia [Irving and Irving, 1982] or China [Lin et al., 1985] are close to the present geographic north pole. As the paleomagnetic inclinations of the Kani Group are consistent with that expected from the site latitude (54.9° at 35.4°N), the declination shift of about 50° should be attributed to tectonic rotation around a pivot close to the rotated block.

Such a declination shift may represent rotation of structural domains of various size between strike-slip faults [e.g., Beck, 1976; Nur et al., 1986]. Kanaori [1990] claimed that the inner belt of Southwest Japan, north of the Median Tectonic Line, can be divided into ten or more small blocks, suggesting that these blocks might be rotated relative to each other. Kano et al. [1990] pointed out the existence of megakinks, i.e. the bending of geological structure at about a 100 km scale, in the pre-Neogene terranes of Southwest Japan. The differential rotation within Southwest Japan, however, is suggested to be of small amount, if any, as the paleomagnetic directions reported between the Mizunami and San'in area show no significant rotation between them. As also shown by Kano et al. [1990] and Kanaori [1990], most of the clockwise deflection of the paleomagnetic directions can be attributed to rotation of Southwest Japan in the dimension of micro-plate size linked to the opening of the Japan Sea.

Our data from the Kani Group reveal no detectable rotation of Southwest Japan between 22 and 16 Ma. The timing of the rotation should be assigned after 16 Ma, as suggested by previous works [Otofuji et al., 1985; Hayashida, 1986]. This assignment is concordant with K-Ar and paleomagnetic data of Miocene volcanic rocks in the San'in area (Figure 1) recently reported by Otofuji et al. [1991]; they observed a rotation of about 40° between 16.1 and 14.3 Ma. While Nakajima et al. [1990] suggested the possibility of rotation by about 14° before 17 Ma, their data do not seem to conflict with our conclusion that the rotation of Southwest Japan occurred *mostly* between 16 and 14 Ma.

How can be the timing of rotation of Southwest Japan between 16 and 14 Ma correlated with the earlier formation of the Japan Sea from 19 to 14 Ma as suggested by the ODP Leg 127 results [Tamaki et al., 1990]? A possible solution is a parallel drifting of Southwest Japan before 16 Ma. The Japan Basin in the northern part of the Japan Sea was estimated to be older than the Yamato Basin [Tamaki, 1986], and is assumed to have been opened before 16 Ma [e.g., Isezaki, 1986]. Fitting of the Kita-Yamato Bank to the 200 to 2000 m isobath of the Asian continental margin suggests parallel opening of the western part of the Japan Basin, by which southeastward translation of Southwest Japan could be caused (Figure 5). In this stage, initial spreading between the Yamato Bank and the Japanese Islands may be assumed, so as to explain the ages of dredged volcanic rocks [Kaneoka et al., 1990] or the ODP results from the Yamato Basin [Tamaki et al., 1990]. The southeastward translation of Southwest Japan of about a few degrees of latitude is not detected by the present data of paleomagnetic inclinations.

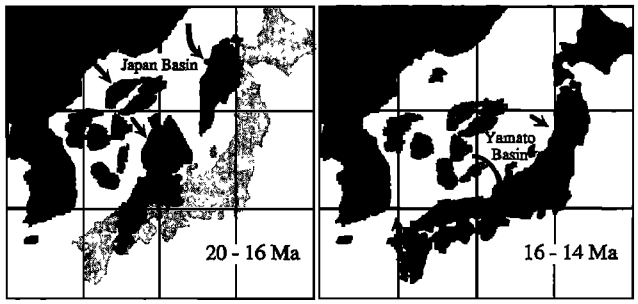



Fig. 5. A possible process for the drift of Southwest Japan associated with the opening of the Japan Sea, comprising southward translation in the interval 20-16 Ma (left) and clockwise rotation in 16-14 Ma (right).

Recent paleomagnetic studies in Northeast Japan suggest that counter-clockwise rotation of Northeastern part of the Japanese Island preceded the clockwise rotation of Southwest Japan [e.g., Yamazaki, 1989]. These data also suggest a possibility of simultaneous drift of Southwest Japan before 16 Ma. It should be assumed that the pre-16 Ma drift of Southwest Japan was mainly a translation, as no significant rotation is detected before 16 Ma.

The clockwise rotation of Southwest Japan between 16 and 14 Ma is probably explained by extension in the southwestern part of the Japan Sea including the Yamato Basin (Figure 5). Assuming the southward drift of Southwest Japan before 16 Ma, we should place the pole of the clockwise rotation inside Southwest Japan, not to the west as suggested by Otofuji and Matsuda [1983]. Our model implies that convergence between the western end of Southwest Japan and the Korean Peninsula occurred around 15 Ma, and may explain the counter-clockwise deflection of paleomagnetic directions observed on the Tsushima Islands and their surroundings [Ishikawa et al., 1989].

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